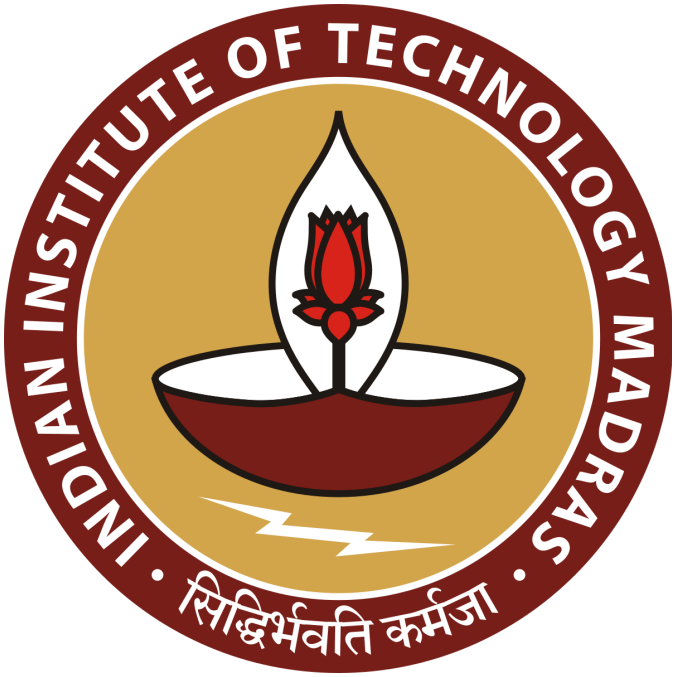
Multi-Dimensional Business Ecosystem Analysis of Tata Motors

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled **"Multi-Dimensional Business Ecosystem Analysis of Tata Motors"**. I extend my appreciation to Tata Motors, for offering the necessary resources through their official website and publicly available reports that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through secondary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.



Signature of Candidate: (Digital Signature)
Name: Raajameenakshi AL
Date: 25/06/2025

1 Executive summary

This project aims to conduct a comprehensive, multi-year, and multi-dimensional analysis of Tata Motors using publicly available data from official sources such as annual reports, BRSR (Business Responsibility and Sustainability Report) and quarterly financial statements. The analysis will evaluate the company's performance across key dimensions including revenue growth, profitability trends, financial resilience, operational health, supply chain stability, and geographical performance. It will also incorporate segment-wise insights, seasonal demand pattern analysis across multiple economic cycles, and comparisons with industry averages. The project will ultimately culminate in the development of a custom Ecosystem Resilience Index (ERI), a composite indicator that quantifies Tata Motors capacity to withstand market disruptions and adapt to evolving business conditions.

2 Organisational Background

Tata Motors Limited, established in 1945, stands as India's automotive manufacturing giant and a subsidiary of the Tata Group conglomerate. Headquartered in Mumbai, the company operates across multiple segments including passenger vehicles, commercial vehicles, and luxury automotive through its Jaguar Land Rover (JLR) division.

- Headquarters: Mumbai,India
- Product Portfolio: Commercial vehicles, passenger vehicles, electric vehicles, luxury vehicles (JLR)
- Manufacturing Facilities: India, UK, China, Brazil, South Africa, and Thailand
- Global Market Presence: Operations in over 125 countries
- Employee Strength: More than 190,000 employees worldwide
- Commercial Vehicles: Market leader in India

3 Problem Statement

3.1 Business Resilience Assessment through Public Disclosures - To develop a structured, index-based framework for evaluating Tata Motor's business resilience by leveraging its publicly available annual and quarterly reports. This framework is intended to unify financial, operational, geographic, and supply chain indicators into a single Business Resilience Index. By doing so, the objective is to systematically assess the company's adaptability over time, uncover hidden patterns, and support data-driven strategic decisions in a competitive and uncertain market environment.

3.2 Trend Analysis for Strategic Planning: To analyze multi-year patterns in financial performance, geographic revenue, and supply chain-linked fluctuations using disclosed company

data. This objective aims to uncover trends that can support proactive decision-making, improve operational adaptability, and strengthen strategic planning under dynamic market conditions.

4 Background of the Problem

Business resilience has emerged as a critical area of interest, especially in industries like automotive, where companies face fluctuations in demand, regulatory changes, evolving technologies, and supply chain complexities. For a company like Tata Motors, resilience is shaped not just by profitability, but also by how well it adapts financially, operationally, and geographically over time.

Internally, factors such as investment decisions, cash flow cycles, and segment profitability trends influence resilience. Externally, global demand shifts, supply chain disruptions (e.g., semiconductor shortages), and regional dependencies (e.g., exposure to the UK through JLR) further affect the company's ability to sustain performance.

While Tata Motors provides detailed annual and quarterly reports, sustainability disclosures, and ESG statements, there is currently no unified framework such as an Business Resilience Index that integrates these data points to assess the company's long-term resilience. Evaluating Tata Motors' performance through a composite index can provide a comprehensive view of its ability to withstand market disruptions across financial, operational, and strategic dimensions. In addition to this, factors like seasonal revenue patterns, geographic sales distribution, and segment-wise dynamics are often reported but not systematically analyzed, leaving key business insights underexplored from a business data management perspective.

5 Problem Solving Approach

- The project will establish a structured approach to assess Tata Motors' resilience by developing a performance-based Business Resilience Index. This index will capture key dimensions such as financial strength, operational stability, and aspects like innovation and sustainability, using only data from official company reports.

- Relevant data will be extracted from Tata Motors' official disclosures such as annual and quarterly reports.The data will be compiled into a structured dataset for further analysis.

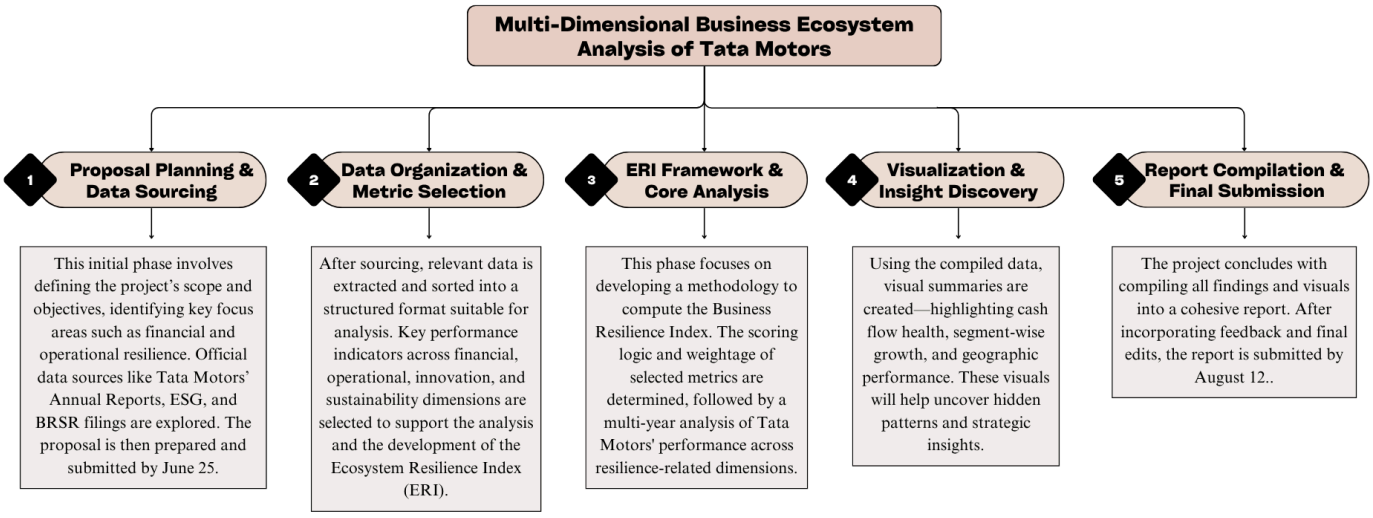
- An Ecosystem Resilience Index (ERI) will be created by selecting appropriate metrics from each key area, standardizing them for comparability, and combining them into a single score that reflects the company's ability to withstand and adapt to business challenges over time.

- Supplementary evaluations such as multi-year financial trend analysis, segment-wise performance review, and geographic revenue breakdown will be performed. Selected indicators will be compared with industry averages where available, to further offer deeper insights

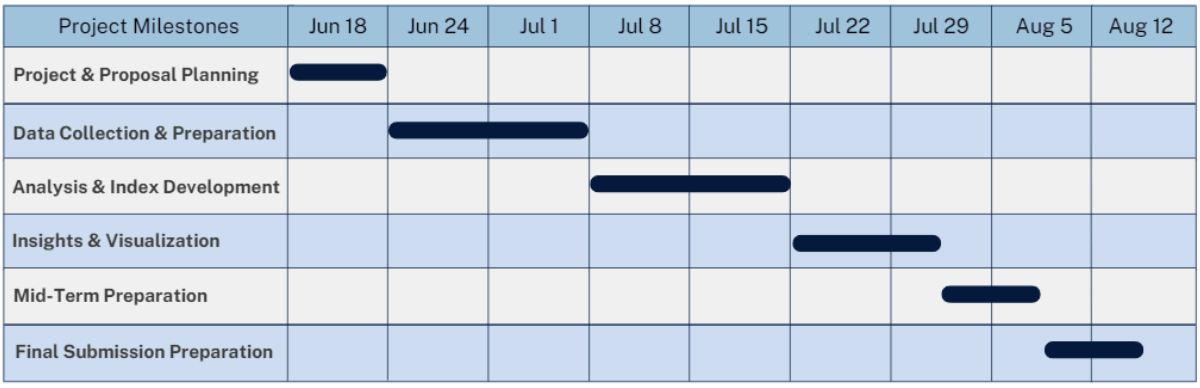
- Key trends and patterns will be visualized using clear and intuitive graphs, especially to highlight cash flow performance, regional growth shifts, and segment contributions.

6 Expected Timeline

Work Breakdown Structure:



Gantt Chart:



7 Expected Outcome

Development of a Unified Evaluation Framework - Business Resilience Index
A comprehensive, data-backed index that consolidates diverse performance indicators, spanning financial strength, operational efficiency, innovation efforts, and sustainability practices into a single, interpretable score. This index will provide an overarching view of Tata Motors' ability to navigate disruptions, maintain stability, and adapt over time.

Deeper Strategic and Operational Insights

- **Multi-Year Performance Trends:**
  Identification of patterns in core financial metrics like profitability, cash flow, and investment behavior, offering a timeline-based understanding of business health.

- **Geographic and Segment-Level Analysis:**
  Evaluation of regional revenue contributions and vehicle segment performance to highlight diversification, dependency, and growth opportunities.

- **Seasonality and Business Cycles:**
  Detection of cyclical patterns using quarterly data to understand demand fluctuations and their implications for strategic planning.